

WT-EXPERIMENT: 06

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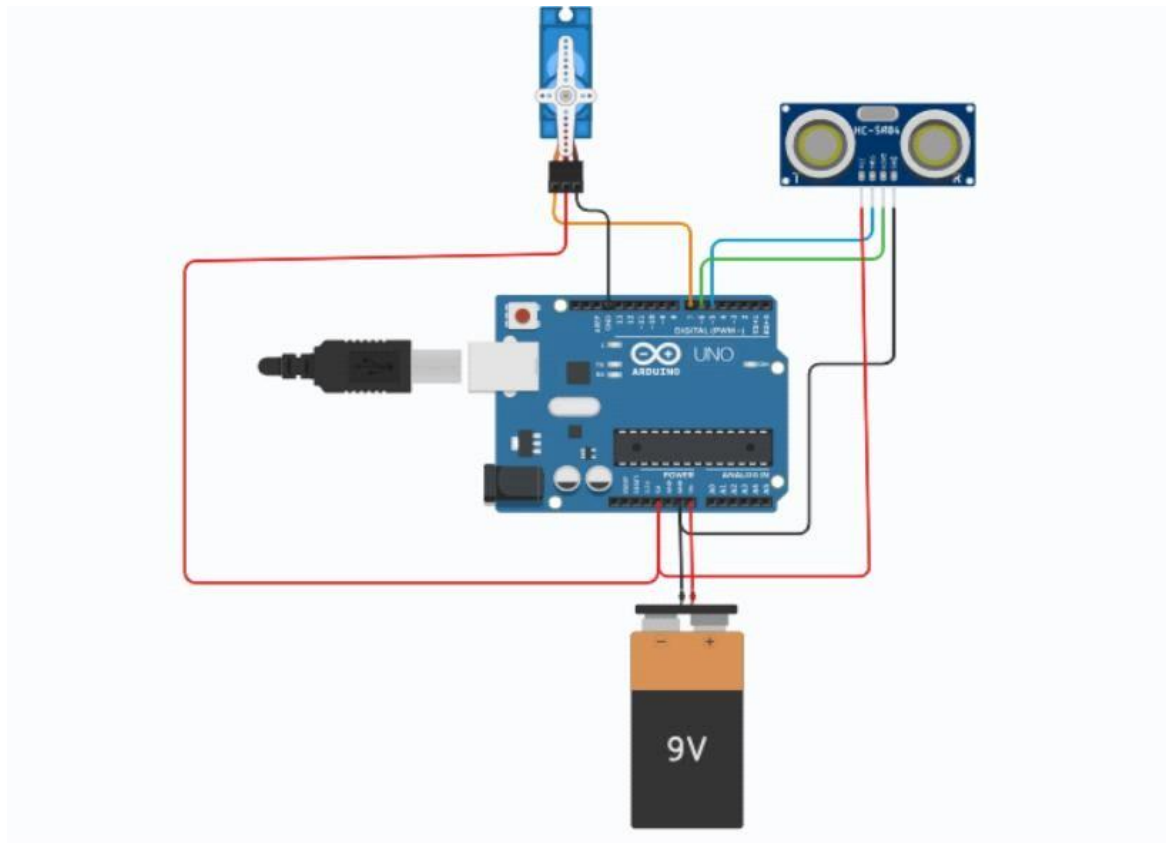
Topic: Automatic Water Dispenser

Aim: To design and implement an automatic water dispenser system using Arduino UNO, an ultrasonic sensor (HC-SR04), and a servo motor to simulate the opening and closing of the tap.

Introduction:

Automatic water dispensers are a modern and hygienic solution for contactless liquid dispensing, commonly found in public restrooms, kitchens, and hospitals. This project simulates the working of such a system using an Arduino UNO, an HC-SR04 ultrasonic sensor, and a servo motor. The aim is to create a touch-free mechanism where water is dispensed when a hand or object is detected within a predefined range.

The ultrasonic sensor is used to detect the presence of an object (such as a hand) by emitting ultrasonic waves and measuring the time it takes for the echo to return. If an object is detected within a short range (e.g., less than 10 cm), the servo motor activates, mimicking the motion of opening a tap or valve to dispense water. After a brief period, the servo returns to its original position, effectively closing the tap.



Components Used:

- Arduino UNO
- HC-SR04 Ultrasonic Sensor
- Servo Motor (SG90 or similar)
- 9V Battery with clip
- Jumper wires
- Breadboard (optional)

Connection of Components with Arduino UNO- Radar Detection System:

Component	Pin on Component	Connected to Arduino UNO
HC-SR04 Ultrasonic Sensor	VCC	5V
	GND	GND
	Trig	Digital Pin 10
	Echo	Digital Pin 11
Servo Motor	Signal (Orange/Yellow)	Digital Pin 9
	VCC (Red)	5V
	GND (Brown/Black)	GND
Power Supply	9V Battery (via clip)	VIN and GND on Arduino
USB Cable (Optional)	USB Port	For code upload and monitoring

Code for Connection:

```
#include <Servo.h> #define
trigPin 10
#define echoPin 11
Servo myServo; void
setup() {
Serial.begin(9600);
myServo.attach(9);
pinMode(trigPin, OUTPUT);
```

```

pinMode(echoPin, INPUT);

}

void loop() {

  for (int angle = 0; angle <= 180; angle += 10) { myServo.write(angle);

    delay(300);

    float distance = measureDistance();

    Serial.print("Angle: ");

    Serial.print(angle);

    Serial.print("° | Distance: ");

    Serial.print(distance); Serial.println("

    cm");

  }

  for (int angle = 180; angle >= 0; angle -= 10) { myServo.write(angle);

    delay(300);

    float distance = measureDistance();

    Serial.print("Angle: ");

    Serial.print(angle);

    Serial.print("° | Distance: ");

    Serial.print(distance); Serial.println("

    cm");

  }

}

float measureDistance() {

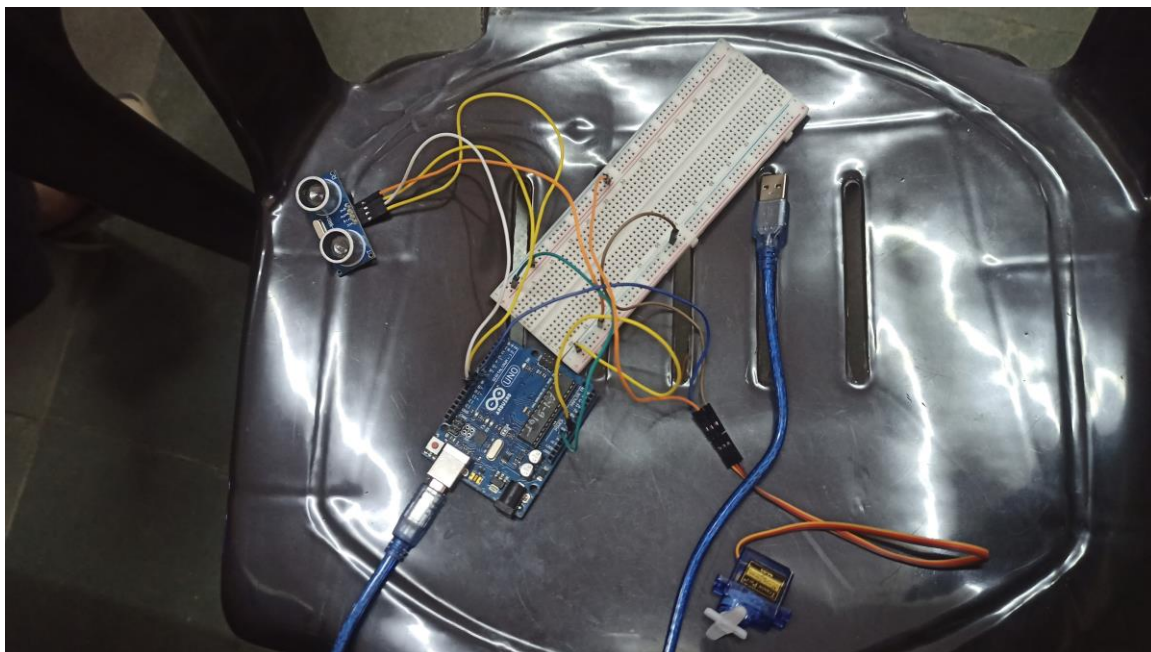
```

```
digitalWrite(trigPin, LOW);  
  
delayMicroseconds(2);  
  
digitalWrite(trigPin, HIGH);  
  
delayMicroseconds(10);  
  
digitalWrite(trigPin, LOW);  
  
long duration = pulseIn(echoPin, HIGH);  
  
float distance = duration * 0.034 / 2; return  
  
distance;  
  
}
```

Libraries Used:

- Servo.h: Allows control over the servo motor to rotate between 0–180° angles.
- No additional libraries are needed as the ultrasonic sensor uses simple digital pins.

Implementation:



- Connect the components according to the wiring diagram (as shown in your image).
- Upload the code to the Arduino UNO via USB.
- Open the Serial Monitor (set baud rate to 9600).
- Place your hand near the sensor (within 10 cm range).
- When a hand is detected, the servo motor rotates to "open" position to simulate dispensing water. After a short delay, it returns to the "closed" position.
- The distance measurements and status updates (e.g., "Hand Detected!") are printed to the Serial Monitor for debugging or testing purposes.

Conclusion:

This project successfully demonstrates a simple and effective automatic water dispenser system using Arduino. By utilizing an ultrasonic sensor to detect the presence of a hand and a servo motor to simulate the opening and closing of a tap, the system provides a contactless and hygienic way to dispense water. It showcases the practical application of sensor-based automation in real-life scenarios such as public restrooms, kitchens, hospitals, and smart homes.